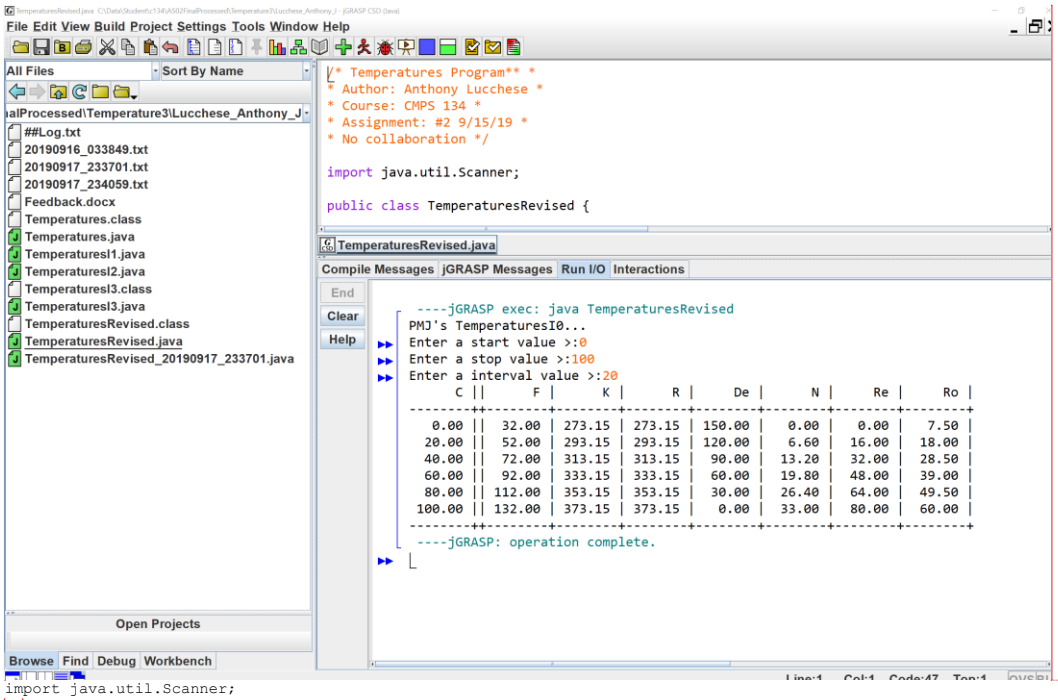
Assignment 2

```
Termperature progression homework
Lucchese_Anthony_J, Mon Sep 16, 2019, 03:38:49
--------------------------------------------------
no message

Files:
  TemperaturesI1.java
  TemperaturesI2.java
  TemperaturesI3.java
  Temperatures.java

revised version of Temperature.java
Lucchese_Anthony_J, Tue Sep 17, 2019, 23:37:01
--------------------------------------------------
decomposed it more thoroughly  but feels unnecessary

Files:
  TemperaturesRevised.java

revised again  forgot something
Lucchese_Anthony_J, Tue Sep 17, 2019, 23:40:59
--------------------------------------------------
no message

Files:
  TemperaturesRevised.java
```




```
import java.util.Scanner;

public class Temperatures {

   static Scanner keyboard = new Scanner(System.in);
   static double celsius, fahrenheit, kelvin, rankine, delisle, newton, reaumur, romer;

   public static void main(String[] args) {
      System.out.println("PMJ's TemperaturesI0...");

      System.out.print("Enter a start value >:");
      double start = keyboard.nextDouble();
      System.out.print("Enter a stop value >:");
      double celsiusCieling = keyboard.nextDouble();
      System.out.print("Enter a interval value >:");
      double increment = keyboard.nextDouble();
      String boundary = "--------++--------+--------+--------+--------+--------+--------+--------+";
      System.out.println("      C ||      F |      K |      R |      De |      N |      Re |      Ro |\n" +
boundary);

      for(double startpoint = celsius ; start <= celsiusCieling; start = start + increment){
      celsius = celsius + increment;
      fahrenheit = celsius * (9/5) + 32;
      kelvin = celsius + 273.15;
      rankine = kelvin * (9/5);
      delisle = (100 - celsius) * 3/2;
      newton = celsius * 33/100;
      reaumur = celsius * 4/5;
      romer = celsius * 21/40 + 7.5;
      System.out.printf("%7.2f ||", celsius);
      printReal(fahrenheit);
      printReal(kelvin);
      printReal(rankine);
      printReal(delisle);
      printReal(newton);
      printReal(reaumur);
```


Commented [PMJ1]: Some of the comments I've made are addressed in this revised version you submitted, but even this version suffers the major flaw. I recognize and applaud your efforts but you also need to focus on the specification and correctness too. Keep at it.

See me for clarification.

C+

Commented [PMJ2]: Required comments missing.

Commented [PMJ3]: The intent was that you would change this to your name or initials!

Commented [PMJ4]: Note the inconsistency here!

Commented [PMJ5]: There should be no need for this statement!

Commented [PMJ6]: Each of these conversions is more properly in a separate method; remember decomposition.

```
      printReal(romer);
      System.out.println();
      }
      System.out.print(boundary);
   }
    public static void printReal(double real) {
         System.out.printf("%7.2f |",real);
   }

}
```


Assignment 2

Assignment 2

```
PMJ's TemperaturesI0...
Enter a start value >:0
Enter a stop value >:100
Enter a interval value >:20
      C ||      F |      K |      R |     De |      N |     Re |     Ro |
--------++--------+--------+--------+--------+--------+--------+--------+
  20.00 ||  52.00 | 293.15 | 293.15 | 120.00 |   6.60 |  16.00 |  18.00 |
  40.00 ||  72.00 | 313.15 | 313.15 |  90.00 |  13.20 |  32.00 |  28.50 |
  60.00 ||  92.00 | 333.15 | 333.15 |  60.00 |  19.80 |  48.00 |  39.00 |
  80.00 || 112.00 | 353.15 | 353.15 |  30.00 |  26.40 |  64.00 |  49.50 |
 100.00 || 132.00 | 373.15 | 373.15 |   0.00 |  33.00 |  80.00 |  60.00 |
 120.00 || 152.00 | 393.15 | 393.15 | -30.00 |  39.60 |  96.00 |  70.50 |
--------++--------+--------+--------+--------+--------+--------+--------+

PMJ's TemperaturesI0...
Enter a start value >:-100
Enter a stop value >:200
Enter a interval value >:50
      C ||      F |      K |      R |     De |      N |     Re |     Ro |
--------++--------+--------+--------+--------+--------+--------+--------+
  50.00 ||  82.00 | 323.15 | 323.15 |  75.00 |  16.50 |  40.00 |  33.75 |
 100.00 || 132.00 | 373.15 | 373.15 |   0.00 |  33.00 |  80.00 |  60.00 |
 150.00 || 182.00 | 423.15 | 423.15 | -75.00 |  49.50 | 120.00 |  86.25 |
 200.00 || 232.00 | 473.15 | 473.15 |-150.00 |  66.00 | 160.00 | 112.50 |
 250.00 || 282.00 | 523.15 | 523.15 |-225.00 |  82.50 | 200.00 | 138.75 |
 300.00 || 332.00 | 573.15 | 573.15 |-300.00 |  99.00 | 240.00 | 165.00 |
 350.00 || 382.00 | 623.15 | 623.15 |-375.00 | 115.50 | 280.00 | 191.25 |
--------++--------+--------+--------+--------+--------+--------+--------+

PMJ's TemperaturesI0...
Enter a start value >:0
Enter a stop value >:100
Enter a interval value >:17
      C ||      F |      K |      R |     De |      N |     Re |     Ro |
--------++--------+--------+--------+--------+--------+--------+--------+
  17.00 ||  49.00 | 290.15 | 290.15 | 124.50 |   5.61 |  13.60 |  16.43 |
  34.00 ||  66.00 | 307.15 | 307.15 |  99.00 |  11.22 |  27.20 |  25.35 |
  51.00 ||  83.00 | 324.15 | 324.15 |  73.50 |  16.83 |  40.80 |  34.28 |
  68.00 || 100.00 | 341.15 | 341.15 |  48.00 |  22.44 |  54.40 |  43.20 |
  85.00 || 117.00 | 358.15 | 358.15 |  22.50 |  28.05 |  68.00 |  52.13 |
 102.00 || 134.00 | 375.15 | 375.15 |  -3.00 |  33.66 |  81.60 |  61.05 |
--------++--------+--------+--------+--------+--------+--------+--------+

PMJ's TemperaturesI0...
Enter a start value >:20
Enter a stop value >:30
Enter a interval value >:1
      C ||      F |      K |      R |     De |      N |     Re |     Ro |
--------++--------+--------+--------+--------+--------+--------+--------+
   1.00 ||  33.00 | 274.15 | 274.15 | 148.50 |   0.33 |   0.80 |   8.03 |
   2.00 ||  34.00 | 275.15 | 275.15 | 147.00 |   0.66 |   1.60 |   8.55 |
   3.00 ||  35.00 | 276.15 | 276.15 | 145.50 |   0.99 |   2.40 |   9.08 |
   4.00 ||  36.00 | 277.15 | 277.15 | 144.00 |   1.32 |   3.20 |   9.60 |
   5.00 ||  37.00 | 278.15 | 278.15 | 142.50 |   1.65 |   4.00 |  10.13 |
   6.00 ||  38.00 | 279.15 | 279.15 | 141.00 |   1.98 |   4.80 |  10.65 |
   7.00 ||  39.00 | 280.15 | 280.15 | 139.50 |   2.31 |   5.60 |  11.18 |
   8.00 ||  40.00 | 281.15 | 281.15 | 138.00 |   2.64 |   6.40 |  11.70 |
   9.00 ||  41.00 | 282.15 | 282.15 | 136.50 |   2.97 |   7.20 |  12.23 |
  10.00 ||  42.00 | 283.15 | 283.15 | 135.00 |   3.30 |   8.00 |  12.75 |
  11.00 ||  43.00 | 284.15 | 284.15 | 133.50 |   3.63 |   8.80 |  13.28 |
--------++--------+--------+--------+--------+--------+--------+--------+

PMJ's TemperaturesI0...
Enter a start value >:30
Enter a stop value >:20
Enter a interval value >:1
      C ||      F |      K |      R |     De |      N |     Re |     Ro |
--------++--------+--------+--------+--------+--------+--------+--------+
--------++--------+--------+--------+--------+--------+--------+--------+
```

Commented [PMJ7]: Misleading title.

Commented [PMJ8]: Do you not realize that your results are incorrect. This, of course, is a major flaw.

Assignment 2

```
import java.util.Scanner;

public class TemperaturesI1 {

   static Scanner keyboard = new Scanner(System.in);
   static double celsius, fahrenheit, kelvin, rankine, delisle, newton, reaumur, romer;

   public static void main(String[] args) {
      System.out.println("PMJ's TemperaturesI0...");

      System.out.print("Enter a Celsius value >:");
      celsius = keyboard.nextDouble();
      fahrenheit = celsius * (9/5) + 32;
      kelvin = celsius + 273.15;
      rankine = kelvin * (9/5);
      delisle = (100 - celsius) * 3/2;
      newton = celsius * 33/100;
      reaumur = celsius * 4/5;
      romer = celsius * 21/40 + 7.5;

      System.out.print(celsius + " || ");
      System.out.print(fahrenheit + " | ");
      System.out.print(kelvin + " | ");
      System.out.print(rankine + " | ");
      System.out.print(delisle + " | ");
      System.out.print(newton + " | ");
      System.out.print(reaumur + " | ");
      System.out.print(romer + " | ");



      System.out.println();

   }

}
```

Assignment 2

```
import java.util.Scanner;

public class TemperaturesI2 {

   static Scanner keyboard = new Scanner(System.in);
   static double fahrenheit, kelvin, rankine, delisle, newton, reaumur, romer;

   public static void main(String[] args) {
      System.out.println("PMJ's TemperaturesI0...");

      System.out.print("Enter a start value >:");
      double celsius = keyboard.nextDouble();
      System.out.print("Enter a stop value >:");
      double celsiusCieling = keyboard.nextDouble();
      System.out.print("Enter a interval value >:");
      double increment = keyboard.nextDouble();


      for(double start = celsius ; celsius <= celsiusCieling; celsius = celsius + increment){
      fahrenheit = celsius * (9/5) + 32;
      kelvin = celsius + 273.15;
      rankine = kelvin * (9/5);
      delisle = (100 - celsius) * 3/2;
      newton = celsius * 33/100;
      reaumur = celsius * 4/5;
      romer = celsius * 21/40 + 7.5;

      System.out.print(celsius + " || ");
      System.out.print(fahrenheit + " | ");
      System.out.print(kelvin + " | ");
      System.out.print(rankine + " | ");
      System.out.print(delisle + " | ");
      System.out.print(newton + " | ");
      System.out.print(reaumur + " | ");
      System.out.print(romer + "\n");}

   }

}
```

Assignment 2

```
import java.util.Scanner;

public class TemperaturesI3 {

   static Scanner keyboard = new Scanner(System.in);
   static double celsius, fahrenheit, kelvin, rankine, delisle, newton, reaumur, romer;

   public static void main(String[] args) {
      System.out.println("PMJ's TemperaturesI0...");

      System.out.print("Enter a start value >:");
      double start = keyboard.nextDouble();
      System.out.print("Enter a stop value >:");
      double celsiusCieling = keyboard.nextDouble();
      System.out.print("Enter a interval value >:");
      double increment = keyboard.nextDouble();
      System.out.println("      C ||      F |      K |      R |     De |      N |     Re |     Ro |");

      for(double startpoint = celsius ; start <= celsiusCieling; start = start + increment){
      fahrenheit = celsius * (9/5) + 32;
      kelvin = celsius + 273.15;
      rankine = kelvin * (9/5);
      delisle = (100 - celsius) * 3/2;
      newton = celsius * 33/100;
      reaumur = celsius * 4/5;
      romer = celsius * 21/40 + 7.5;
      System.out.printf("%7.2f ||", celsius);
      printReal(fahrenheit);
      printReal(kelvin);
      printReal(rankine);
      printReal(delisle);
      printReal(newton);
      printReal(reaumur);
      printReal(romer);
      System.out.println();
      }
   }
    public static void printReal(double real) {
         System.out.printf("%7.2f |",real);
   }

}
```

Assignment 2